

Method Selection Guide

K-Means

Assumption → Clusters are reasonably compact and approximately spherical.

Good At → Fast segmentation of compact, well-separated groups when K is known.

Fails When → Clusters are strongly curved or have unusual shapes.

Ward Hierarchical Clustering

Assumption → Clusters should be compact and merging should increase within cluster variance as little as possible.

Good At → Compact, well-separated groups and hierarchical exploration.

Fails When → The actual structure is strongly curved or irregular.

Single Linkage

Assumption → Nearby observations can be used to connect observations into groups.

Good At → Curved and elongated structures.

Fails When → A thin bridge connects two otherwise separate groups because of the chaining effect.

DBSCAN

Assumption → Clusters are dense regions separated by lower-density areas.

Good At → Irregular shapes and identifying noise/outliers.

Fails When → Cluster densities differ strongly or eps is poorly chosen.

GMM

Assumption → Data can be represented using a mixture of probability distributions.

Good At → Soft membership, uncertainty and stretched/elliptical clusters.

Fails When → The assumed distribution does not represent the actual shape well.

Decision Guide

- ⑩ If the data contains compact, approximately spherical groups: **use K-Means.**
- ⑩ If compact groups are present and a hierarchical structure is useful: **try Ward hierarchical clustering.**
- ⑩ If the clusters are curved or connected: **try Single Linkage, while checking for chaining.**
- ⑩ If the data contains dense irregular groups and possible outliers: **try DBSCAN.**
- ⑩ If uncertainty in cluster membership matters or clusters are stretched: **try GMM.**
- ⑩ If the number of GMM components is unclear: **compare BIC across several K values.**

Three worked Examples

1. Blobs →

```
blob_scores = table.loc[
    "Blobs"
]
```

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```
print(
    blob_scores.sort_values(
        ascending=False
    )
)
```

```
output :-
k-means      1.000
ward         1.000
single       1.000
GMM          1.000
DBSCAN (eps=0.2) 0.997
Name: Blobs, dtype: float64
```

- ⑩ The blobs dataset contains compact, separated groups. The ARI table shows how each method performed, and the method with the highest ARI is the strongest match for this particular shape.

2. Moons →

```
moon_scores = table.loc[
    "Moons"
]

print(
    moon_scores.sort_values(
        ascending=False
    )
)
```

```
output :-
single      1.000
DBSCAN (eps=0.2) 0.984
ward        0.627
GMM         0.461
k-means     0.445
Name: Moons, dtype: float64
```

- ⑩ The moons dataset contains curved clusters rather than compact spherical groups. The bake-off results show that methods based on connectivity or density can capture this structure better than methods that depend mainly on cluster centres.

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3. Customer RFM →

```
print(  
    "GMM BIC selected K:",  
    best_k  
)
```

```
print(  
    "Day 31 business K:",  
    day31_k  
)
```

```
print(  
    "DBSCAN clusters:",  
    n_dbscan_clusters  
)
```

```
print(  
    "DBSCAN noise points:",  
    n_noise  
)
```

output :-

GMM BIC selected K: 5

Day 31 Business K: 5

DBSCAN clusters: 1

DBSCAN noise points: 16

- ⑩ For the customer RFM data, GMM selected the number of components by minimising BIC, while Day 31 selected K=5 for business reasons. Comparing the two provides evidence about whether the business segmentation is supported by the statistical structure. DBSCAN answers a different question because it identifies dense regions and noise rather than forcing customers into a chosen number of business segments.

One situation where I would try two methods →

If I received a new customer dataset and did not know whether the customers formed compact groups or irregular dense regions, I would try GMM and DBSCAN.

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GMM would provide cluster membership probabilities and identify uncertain customers, while DBSCAN would test for dense regions and potentially unusual observations. I would compare both the numerical results and the business meaning of the resulting groups before choosing an approach.

Final Conclusion →

There is no single best clustering method. The experiments show that different methods succeed under different assumptions about cluster shape. K-Means and Ward are suitable for compact groups, Single Linkage and DBSCAN can capture curved structures, and GMM can model stretched groups and provide membership probabilities.

Therefore, the correct approach is to first understand the shape of the data and the question being asked, then choose and validate a method rather than assuming one clustering algorithm will work for every dataset.